

Matematisk statistik

KTH

Formelsamling för statistisk inlärning och dataanalys

1 Notation och specialfunktioner

För ett positivt heltal n sätter vi $[n] = \{1, 2, \dots, n\}$.

För positiva heltal n och k har vi binomialkoefficienten $\binom{n}{k} = \begin{cases} \frac{n!}{k!(n-k)!} & 0 \leq k \leq n, \\ 0 & \text{annars.} \end{cases}$

Gammafunktionen $\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t} dt$ med egenskaperna $\Gamma(1) = 1$, $\Gamma(x+1) = x\Gamma(x)$ för $x > 0$.

För ett positivt heltal k har vi $\Gamma(k) = (k-1)!$.

2 Endimensionella fördelningar

Beteckning	Parametrar	Stöd	$f_X(x)$	$E[X]$	$\text{Var}[X]$
Ber(p)	$p \in [0, 1]$	$\{0, 1\}$	$p^x(1-p)^{1-x}$	p	$p(1-p)$
Bin(n, p)	$n \in \mathbb{Z}_{>0}$ $p \in [0, 1]$	$\mathbb{Z}_{\geq 0}$ $x \leq n$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Geom(p)	$p \in [0, 1]$	$\mathbb{Z}_{\geq 0}$	$p(1-p)^x$	$(1-p)/p$	$(1-p)/p^2$
NegBin(r, p)	$r \in \mathbb{Z}_{>0}$ $p \in [0, 1]$	$\mathbb{Z}_{\geq 0}$	$\binom{r+x-1}{x} p^r (1-p)^x$	$r(1-p)/p$	$r(1-p)/p^2$
Po(λ)	$\lambda \in \mathbb{R}_{>0}$	$\mathbb{Z}_{\geq 0}$	$\frac{e^{-\lambda} \lambda^x}{x!}$	λ	λ
U(a, b)	$a, b \in \mathbb{R}$ $a < b$	$[a, b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Beta(α, β)	$\alpha, \beta > 0$	$[0, 1]$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
$\chi^2(\nu)$	$\nu \in \mathbb{Z}_{>0}$	$\mathbb{R}_{>0}$	$\frac{1}{\Gamma(\nu/2)2^{\nu/2}} x^{\nu/2-1} e^{-x/2}$	ν	2ν
Exp(λ)	$\lambda \in \mathbb{R}_{>0}$	$\mathbb{R}_{>0}$	$\lambda e^{-\lambda x}$	λ^{-1}	λ^{-2}
Gamma(α, λ)	$\alpha, \lambda \in \mathbb{R}_{>0}$	$\mathbb{R}_{>0}$	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
InvGamma(α, β)	$\alpha, \beta \in \mathbb{R}_{>0}$	$\mathbb{R}_{>0}$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\beta/x}$	$\frac{\beta}{\alpha-1} \ (\alpha > 1)$	$\frac{\beta^2}{(\alpha-1)^2(\alpha-2)} \ (\alpha > 2)$
N(μ, σ^2)	$\mu \in \mathbb{R}$ $\sigma^2 \in \mathbb{R}_{>0}$	$\mathbb{R}$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
t(ν)	$\nu \in \mathbb{Z}_{>0}$	$\mathbb{R}$	$\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	$0 \ (\nu > 1)$	$\frac{\nu}{\nu-2} \ (\nu > 2)$
t(ν, μ, σ^2)	$\nu \in \mathbb{Z}_{>0}$	$\mathbb{R}$	$\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi\sigma^2}} \left(1 + \frac{1}{\nu} \frac{(x-\mu)^2}{\sigma^2}\right)^{-\frac{\nu+1}{2}}$	$\mu \ (\nu > 1)$	$\frac{\nu}{\nu-2} \sigma^2 \ (\nu > 2)$
BetaBin(n, α, β)	$n \in \mathbb{Z}_{\geq 0}$ $\alpha, \beta \in \mathbb{R}_{>0}$	$\mathbb{Z}_{\geq 0}$ $x \leq n$	$\binom{n}{x} \frac{\Gamma(\alpha+x)\Gamma(\beta+n-x)}{\Gamma(\alpha+\beta+n)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$	$\frac{n\alpha}{\alpha+\beta}$	$\frac{n\alpha\beta(\alpha+\beta+n)}{(\alpha+\beta)^2(\alpha+\beta+1)}$
BetaNegBin(r, α, β)	$r \in \mathbb{Z}_{\geq 0}$ $\alpha, \beta \in \mathbb{R}_{>0}$	$\mathbb{Z}_{\geq 0}$	$\binom{r+x-1}{x} \frac{\Gamma(\alpha+r)\Gamma(\beta+x)}{\Gamma(\alpha+\beta+r+x)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$	$\frac{r\beta}{\alpha-1} \ (\alpha > 1)$	$\frac{r(\alpha+r-1)\beta(\alpha+\beta-1)}{(\alpha-2)(\alpha-1)^2} \ (\alpha > 2)$

Om $X \sim \chi^2(2n)$ har vi att $X \sim \text{Gamma}(n, 1/2)$ och därför också att $\frac{X}{2\lambda} \sim \text{Gamma}(n, \lambda)$.

3 Flerdimensionella fördelningar

Beteckning	Parametrar	Stöd	$f_{\mathbf{X}}(\mathbf{x})$	$E[X_i]$	$\text{Cov}[X_i, X_j]$
$N(\boldsymbol{\mu}, \Sigma)$	$\boldsymbol{\mu} \in \mathbb{R}^m$ $\Sigma \in \mathbb{R}^{m \times m}$	$\mathbb{R}^m$	$\frac{1}{(2\pi)^{m/2} \sqrt{\det \Sigma}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$	μ_i	Σ_{ij}
$\text{Multi}(n, \mathbf{p})$	$n \in \mathbb{Z}_{>0}$ $\mathbf{p} \in [0, 1]^m$ $\sum p_i = 1$	$\mathbb{Z}_{>0}^m$ $\sum x_i = n$	$\frac{n!}{x_1! \dots x_m!} p_1^{x_1} \dots p_m^{x_m}$	np_i	$np_i(1-p_i) \quad (i=j)$ $-np_i p_j \quad (i \neq j)$
$\text{Dirichlet}(\boldsymbol{\alpha})$	$\boldsymbol{\alpha} \in \mathbb{R}_{>0}^m$	$[0, 1]^m$ $\sum x_i = 1$	$\frac{\Gamma(\alpha_0)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_m)} \prod_{i=1}^m x_i^{\alpha_i-1}$ där $\alpha_0 = \sum \alpha_i$	$\frac{\alpha_i}{\alpha_0}$	$\frac{\alpha_i(\alpha_0-\alpha_i)}{\alpha_0^2(\alpha_0+1)} \quad (i=j)$ $-\frac{\alpha_i \alpha_j}{\alpha_0^2(\alpha_0+1)} \quad (i \neq j)$
$\text{NInvGamma}(\mu, \lambda, \alpha, \beta)$	$\mu \in \mathbb{R}$ $\lambda, \alpha, \beta \in \mathbb{R}_{>0}$	$\mathbb{R} \times \mathbb{R}_{>0}$	$\frac{\sqrt{\lambda} \beta^\alpha x_V^{\alpha-1}}{(2\pi x_V)^{1/2} \Gamma(\alpha)} e^{-\frac{1}{2x_V}(2\beta + \lambda(x_M - \mu)^2)}$		

4 Hierarkier

$\mathbf{X} \mid \mathbf{Y} = \mathbf{y}$	$\mathbf{Y}$	$\mathbf{X}$
$\text{Bin}(n, y)$	$\text{Beta}(\alpha, \beta)$	$\text{BetaBin}(n, \alpha, \beta)$
$\text{NegBin}(r, y)$	$\text{Beta}(\alpha, \beta)$	$\text{BetaNegBin}(r, \alpha, \beta)$
$\text{Po}(y)$	$\text{Gamma}(\alpha, \lambda)$	$\text{NegBin}(\alpha, \lambda/(\lambda+1))$
$N(y, \sigma^2)$	$N(\mu, \tau^2)$	$N(\mu, \sigma^2 + \tau^2)$
$N(\mu, y)$	$\text{InvGamma}(\alpha, \beta)$	$t(2\alpha, \mu, \beta/\alpha)$
$N(y_M, y_V)$	$\text{NInvGamma}(\mu, \lambda, \alpha, \beta)$	$t(2\alpha, \mu, \beta(\lambda+1)/\alpha\lambda)$
$N(\mathbf{y}, \Sigma)$	$N(\boldsymbol{\mu}, \Lambda)$	$N(\boldsymbol{\mu}, \Sigma + \Lambda)$

5 Exponentialfamiljer

Täthetsfunktionerna (eller sannolikhetsfunktionerna) $f_{\mathbf{X}}(\mathbf{x} \mid \boldsymbol{\theta})$ bildar en *exponentialfamilj* med avseende på parametrerna $\boldsymbol{\theta} \in \Omega$ om $f_{\mathbf{X}}(\mathbf{x} \mid \boldsymbol{\theta})$ kan skrivas som

$$f_{\mathbf{X}}(\mathbf{x} \mid \boldsymbol{\theta}) = h(\mathbf{x})c(\boldsymbol{\theta})e^{\sum_{i=1}^k w_i(\boldsymbol{\theta})^T t_i(\mathbf{x})}$$

för någon $h: \mathbb{R}^m \rightarrow \mathbb{R}_{\geq 0}$, $c: \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ and $w_i: \mathbb{R}^d \rightarrow \mathbb{R}^{k_i}$, $t_i: \mathbb{R}^m \rightarrow \mathbb{R}^{k_i}$ där $k_i > 0$ för $i = 1, \dots, k$.

Om $\mathbf{X}$ har en fördelning från en exponentialfamilj så gäller att

$$\begin{aligned} E \left[\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j}^T t_i(\mathbf{X}) \right] &= -\frac{\partial}{\partial \theta_j} \log c(\boldsymbol{\theta}) \\ \text{Var} \left[\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j}^T t_i(\mathbf{X}) \right] &= -\frac{\partial^2}{\partial \theta_j^2} \log c(\boldsymbol{\theta}) - E \left[\sum_{i=1}^k \frac{\partial^2 w_i(\boldsymbol{\theta})}{\partial \theta_j^2}^T t_i(\mathbf{X}) \right] \end{aligned}$$

6 Marginalfördelningar och betingade fördelningar

Antag att $\mathbf{X} \sim N(\boldsymbol{\mu}, \Sigma)$ för $\boldsymbol{\mu} \in \mathbb{R}^m$ och $\Sigma \in \mathbb{R}^{m \times m}$ positivt definit. Om $\mathbf{X}$, $\boldsymbol{\mu}$ och Σ kan skrivas som

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix} \quad \text{respektive} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

där $\mathbf{X}_1, \boldsymbol{\mu}_1 \in \mathbb{R}^k$ och $\Sigma_{11} \in \mathbb{R}^{k \times k}$, $\Sigma_{22} \in \mathbb{R}^{(m-k) \times (m-k)}$ så har vi

$$\mathbf{X}_1 \sim N(\boldsymbol{\mu}_1, \Sigma_{11}),$$

och

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim N(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}).$$

Om $\mathbf{X} \sim \text{Multi}(n, \mathbf{p})$ så gäller

$$\left[X_1, \dots, X_k, \sum_{i=k+1}^m X_i \right]^T \sim \text{Multi} \left(n, \left[p_1, \dots, p_k, \sum_{i=k+1}^m p_i \right]^T \right)$$

och

$$[X_1, X_2, \dots, X_k]^T \mid [X_{k+1}, \dots, X_m]^T = [x_{k+1}, \dots, x_m]^T \sim \text{Multi} \left(n - \sum_{i=k+1}^m x_i, [p_1, \dots, p_k]^T / \left(1 - \sum_{i=k+1}^m p_i \right) \right).$$

Om $\mathbf{X} \sim \text{Dirichlet}(\boldsymbol{\alpha})$ så gäller

$$\left[X_1, \dots, X_k, \sum_{i=k+1}^m X_i \right]^T \sim \text{Dirichlet} \left(\left[\alpha_1, \dots, \alpha_k, \sum_{i=k+1}^m \alpha_i \right]^T \right)$$

och

$$\frac{[X_1, \dots, X_k]^T}{\sum_{i=1}^k X_i} \mid [X_{k+1}, \dots, X_m]^T = [x_{k+1}, \dots, x_m]^T \sim \text{Dirichlet}([\alpha_1, \dots, \alpha_k]^T)$$

Om $X_M, X_V \sim \text{NInvGamma}(\mu, \lambda, \alpha, \beta)$ så har vi

$$X_M \sim t \left(2\alpha, \mu, \frac{\beta}{\alpha\lambda} \right) \quad \text{och} \quad X_V \sim \text{InvGamma}(\alpha, \beta).$$

7 Punktskattning

Cramér–Raos sats. Låt $X_1, \dots, X_n$ vara ett slumpmässigt stickprov från en exponentialfamilj med täthetsfunktion (eller sannolikhetsfunktion) $f_X(x \mid \theta)$ och låt $W(\mathbf{X})$ vara en väntevärdesriktig skattning av $g(\theta)$. Då gäller

$$\text{Var}[W(\mathbf{X})] \geq \frac{g'(\theta)^2}{I(\theta)}$$

där $I(\theta)$ är *informationstalet* eller *fisherinformationen*

$$I(\theta) = n \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(X \mid \theta) \right)^2 \right] = -n \mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f_X(X \mid \theta) \right] = nJ(\theta),$$

där $J(\theta)$ är det *normaliserade informationstalet* eller den *normaliserade fisherinformationen*. Skattningen $W(\mathbf{X})$ uppnår denna olikhet om och endast om

$$a(\theta)(W(\mathbf{x}) - g(\theta)) = \frac{\partial}{\partial \theta} \log f_{\mathbf{X}}(\mathbf{x} \mid \theta)$$

för någon funktion $a(\theta)$ där $f_{\mathbf{X}}(\mathbf{x} \mid \theta)$ är täthetsfunktionen (eller sannolikhetsfunktionen) för hela stickprovet.

Om $X_1, \dots, X_n$ är oberoende och $N(\mu, \sigma^2)$ -fördelade har vi att

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1) \quad \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1),$$

där $\bar{X} = n^{-1} \sum_{i=1}^n X_i$ och $S^2 = (n-1)^{-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

8 Bayesiansk inferens

Antag att vi har ett stickprov $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ betingat oberoende givet $\boldsymbol{\Theta} = \boldsymbol{\theta}$ med samma datafördelning $\mathbf{X}_i \mid \boldsymbol{\Theta} = \boldsymbol{\theta}$ för $i = 1, \dots, n$. Låt $\bar{x} = n^{-1} \sum_{i=1}^n x_i$ och $s^2 = (n-1)^{-1} \sum_{i=1}^n (x_i - \bar{x})^2$.

Datafördelning	Apriorifördelning	Uppdateringsregel
$\text{Exp}(\theta)$	$\text{Gamma}(\alpha, \lambda)$	$(\alpha, \lambda) \mapsto (\alpha + n, \lambda + n\bar{x})$
$\text{Po}(\theta)$	$\text{Gamma}(\alpha, \lambda)$	$(\alpha, \lambda) \mapsto (\alpha + n\bar{x}, \lambda + n)$
$\text{Bin}(m, \theta)$	$\text{Beta}(\alpha, \beta)$	$(\alpha, \beta) \mapsto (\alpha + n\bar{x}, \beta + nm - n\bar{x})$
$\text{N}(\theta, \sigma^2)$	$\text{N}(\mu, \tau^2)$	$(\mu, \tau^2) \mapsto ((\frac{\mu}{\tau^2} + \frac{n\bar{x}}{\sigma^2}) / (\frac{1}{\tau^2} + \frac{n}{\sigma^2}), 1 / (\frac{1}{\tau^2} + \frac{n}{\sigma^2}))$
$\text{N}(\mu, \theta)$	$\text{InvGamma}(\alpha, \beta)$	$(\alpha, \beta) \mapsto (\alpha + \frac{n}{2}, \beta + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2)$
$\text{Multi}(\boldsymbol{\theta})$	$\text{Dirichlet}(\boldsymbol{\alpha})$	$\boldsymbol{\alpha} \mapsto \boldsymbol{\alpha} + \mathbf{x}$
$\text{N}(\theta_M, \theta_V)$	$\text{NInvGamma}(\mu, \lambda, \alpha, \beta)$	$(\mu, \lambda, \alpha, \beta) \mapsto (\frac{\lambda\mu + n\bar{x}}{\lambda + n}, \lambda + n, \alpha + n/2, \beta + \frac{n-1}{2}s^2 + \frac{n\lambda}{\lambda+n} \frac{(\bar{x}-\mu)^2}{2})$

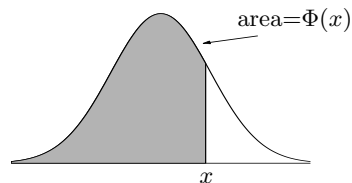
Vi har också följande icke-informativa apriorifördelningar med tillhörande aposteriorifördelningar.

Datafördelning	Apriorifördelning	Aposteriorifördelning
$\text{Bin}(m, \theta)$	$f_{\boldsymbol{\Theta}}(\boldsymbol{\theta}) \propto 1$	$\text{Beta}(1 + n\bar{x}, 1 + nm - n\bar{x})$
$\text{Bin}(m, \theta)$	$f_{\boldsymbol{\Theta}}(\boldsymbol{\theta}) \propto \theta^{-1}(1 - \theta)^{-1}$	$\text{Beta}(n\bar{x}, nm - n\bar{x})$
$\text{N}(\theta, \sigma^2)$	$f_{\boldsymbol{\Theta}}(\boldsymbol{\theta}) \propto 1$	$\text{N}(\bar{x}, \frac{\sigma^2}{n})$
$\text{N}(\mu, \theta)$	$f_{\boldsymbol{\Theta}}(\boldsymbol{\theta}) \propto 1/\theta$	$\text{InvGamma}(\frac{n}{2}, \frac{\sum_{i=1}^n (x_i - \mu)^2}{2})$
$\text{Multi}(\boldsymbol{\theta})$	$f_{\boldsymbol{\Theta}}(\boldsymbol{\theta}) \propto 1$	$\text{Dirichlet}(\mathbf{1} + \mathbf{x})$
$\text{Multi}(\boldsymbol{\theta})$	$f_{\boldsymbol{\Theta}}(\boldsymbol{\theta}) \propto \frac{1}{\theta_1 \theta_2 \dots \theta_m}$	$\text{Dirichlet}(\mathbf{x})$
$\text{N}(\theta_M, \theta_V)$	$f_{\boldsymbol{\Theta}_M, \boldsymbol{\Theta}_V}(\theta_M, \theta_V) \propto 1/\theta_V$	$\text{NInvGamma}(\bar{x}, n, \frac{n-1}{2}, \frac{n-1}{2}s^2)$

Tabell 1. Standard normalfördelning.

$\Phi(x) = P(X \leq x)$, där $X \in N(0, 1)$.

För negativa x , utnyttja att $\Phi(-x) = 1 - \Phi(x)$

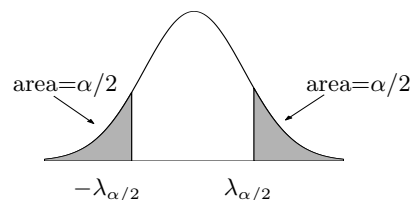
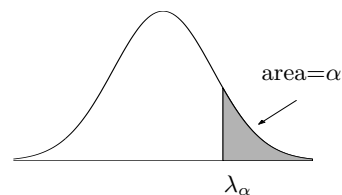


x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865									
3.1	.99903									
3.2	.99931									
3.3	.99952									
3.4	.99966									
3.5	.99977									
3.6	.99984									
3.7	.99989									
3.8	.99993									
3.9	.99995									
4.0	.99997									

Tab 2. Normalfördelningens kvantiler

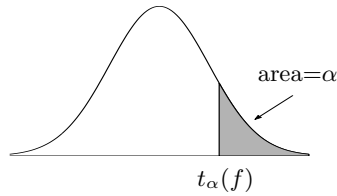
$P(X > \lambda_\alpha) = \alpha$ där $X \in N(0, 1)$

α	λ_α	α	λ_α
0.10	1.2816	0.001	3.0902
0.05	1.6449	0.0005	3.2905
0.025	1.9600	0.0001	3.7190
0.010	2.3263	0.00005	3.8906
0.005	2.5758	0.00001	4.2649



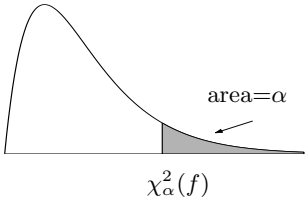
Tabell 3. t -fördelningen.

$P(X > t_\alpha(f)) = \alpha$, där $X \in t(f)$.



f	α	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
1		3.08	6.31	12.71	31.82	63.66	318.31	636.62
2		1.89	2.92	4.30	6.96	9.92	22.33	31.60
3		1.64	2.35	3.18	4.54	5.84	10.21	12.92
4		1.53	2.13	2.78	3.75	4.60	7.17	8.61
5		1.48	2.02	2.57	3.36	4.03	5.89	6.87
6		1.44	1.94	2.45	3.14	3.71	5.21	5.96
7		1.41	1.89	2.36	3.00	3.50	4.79	5.41
8		1.40	1.86	2.31	2.90	3.36	4.50	5.04
9		1.38	1.83	2.26	2.82	3.25	4.30	4.78
10		1.37	1.81	2.23	2.76	3.17	4.14	4.59
11		1.36	1.80	2.20	2.72	3.11	4.02	4.44
12		1.36	1.78	2.18	2.68	3.05	3.93	4.32
13		1.35	1.77	2.16	2.65	3.01	3.85	4.22
14		1.35	1.76	2.14	2.62	2.98	3.79	4.14
15		1.34	1.75	2.13	2.60	2.95	3.73	4.07
16		1.34	1.75	2.12	2.58	2.92	3.69	4.01
17		1.33	1.74	2.11	2.57	2.90	3.65	3.97
18		1.33	1.73	2.10	2.55	2.88	3.61	3.92
19		1.33	1.73	2.09	2.54	2.86	3.58	3.88
20		1.33	1.72	2.09	2.53	2.85	3.55	3.85
21		1.32	1.72	2.08	2.52	2.83	3.53	3.82
22		1.32	1.72	2.07	2.51	2.82	3.50	3.79
23		1.32	1.71	2.07	2.50	2.81	3.48	3.77
24		1.32	1.71	2.06	2.49	2.80	3.47	3.75
25		1.32	1.71	2.06	2.49	2.79	3.45	3.73
26		1.31	1.71	2.06	2.48	2.78	3.43	3.71
27		1.31	1.70	2.05	2.47	2.77	3.42	3.69
28		1.31	1.70	2.05	2.47	2.76	3.41	3.67
29		1.31	1.70	2.05	2.46	2.76	3.40	3.66
30		1.31	1.70	2.04	2.46	2.75	3.39	3.65
40		1.30	1.68	2.02	2.42	2.70	3.31	3.55
60		1.30	1.67	2.00	2.39	2.66	3.23	3.46
120		1.29	1.66	1.98	2.36	2.62	3.16	3.37
∞		1.28	1.64	1.96	2.33	2.58	3.09	3.29

Tabell 4. χ^2 -fördelningen.
 $P(X > \chi^2_\alpha(f)) = \alpha$, där $X \in \chi^2(f)$.



f	α	0.9995	0.999	0.995	0.99	0.975	0.95	0.05	0.025	0.01	0.005	0.001	0.0005
1		0.00	0.00	0.00	0.00	0.00	0.00	3.84	5.02	6.63	7.88	10.8	12.1
2		0.00	0.00	0.01	0.02	0.05	0.10	5.99	7.38	9.21	10.6	13.8	15.2
3		0.02	0.02	0.07	0.11	0.22	0.35	7.81	9.35	11.3	12.8	16.3	17.7
4		0.06	0.09	0.21	0.30	0.48	0.71	9.49	11.1	13.3	14.9	18.5	20.0
5		0.16	0.21	0.41	0.55	0.83	1.15	11.1	12.8	15.1	16.7	20.5	22.1
6		0.30	0.38	0.68	0.87	1.24	1.64	12.6	14.4	16.8	18.5	22.5	24.1
7		0.48	0.60	0.99	1.24	1.69	2.17	14.1	16.0	18.5	20.3	24.3	26.0
8		0.71	0.86	1.34	1.65	2.18	2.73	15.5	17.5	20.1	22.0	26.1	27.9
9		0.97	1.15	1.73	2.09	2.70	3.33	16.9	19.0	21.7	23.6	27.9	29.7
10		1.26	1.48	2.16	2.56	3.25	3.94	18.3	20.5	23.2	25.2	29.6	31.4
11		1.59	1.83	2.60	3.05	3.82	4.57	19.7	21.9	24.7	26.8	31.3	33.1
12		1.93	2.21	3.07	3.57	4.40	5.23	21.0	23.3	26.2	28.3	32.9	34.8
13		2.31	2.62	3.57	4.11	5.01	5.89	22.4	24.7	27.7	29.8	34.5	36.5
14		2.70	3.04	4.07	4.66	5.63	6.57	23.7	26.1	29.1	31.3	36.1	38.1
15		3.11	3.48	4.60	5.23	6.26	7.26	25.0	27.5	30.6	32.8	37.7	39.7
16		3.54	3.94	5.14	5.81	6.91	7.96	26.3	28.8	32.0	34.3	39.3	41.3
17		3.98	4.42	5.70	6.41	7.56	8.67	27.6	30.2	33.4	35.7	40.8	42.9
18		4.44	4.90	6.26	7.01	8.23	9.39	28.9	31.5	34.8	37.2	42.3	44.4
19		4.91	5.41	6.84	7.63	8.91	10.1	30.1	32.9	36.2	38.6	43.8	46.0
20		5.40	5.92	7.43	8.26	9.59	10.9	31.4	34.2	37.6	40.0	45.3	47.5
21		5.90	6.45	8.03	8.90	10.3	11.6	32.7	35.5	38.9	41.4	46.8	49.0
22		6.40	6.98	8.64	9.54	11.0	12.3	33.9	36.8	40.3	42.8	48.3	50.5
23		6.92	7.53	9.26	10.2	11.7	13.1	35.2	38.1	41.6	44.2	49.7	52.0
24		7.45	8.08	9.89	10.9	12.4	13.8	36.4	39.4	43.0	45.6	51.2	53.5
25		7.99	8.65	10.5	11.5	13.1	14.6	37.7	40.6	44.3	46.9	52.6	54.9
26		8.54	9.22	11.2	12.2	13.8	15.4	38.9	41.9	45.6	48.3	54.1	56.4
27		9.09	9.80	11.8	12.9	14.6	16.2	40.1	43.2	47.0	49.6	55.5	57.9
28		9.66	10.4	12.5	13.6	15.3	16.9	41.3	44.5	48.3	51.0	56.9	59.3
29		10.2	11.0	13.1	14.3	16.0	17.7	42.6	45.7	49.6	52.3	58.3	60.7
30		10.8	11.6	13.8	15.0	16.8	18.5	43.8	47.0	50.9	53.7	59.7	62.2
40		16.9	17.9	20.7	22.2	24.4	26.5	55.8	59.3	63.7	66.8	73.4	76.1
50		23.5	24.7	28.0	29.7	32.4	34.8	67.5	71.4	76.2	79.5	86.7	89.6
60		30.3	31.7	35.5	37.5	40.5	43.2	79.1	83.3	88.4	92.0	99.6	103
70		37.5	39.0	43.3	45.4	48.8	51.7	90.5	95.0	100	104	112	116
80		44.8	46.5	51.2	53.5	57.2	60.4	102	107	112	116	125	128
90		52.3	54.2	59.2	61.8	65.6	69.1	113	118	124	128	137	141
100		59.9	61.9	67.3	70.1	74.2	77.9	124	130	136	140	149	153